

UNISOFT SYSTEMS LIMITED

ULMS

Unisoft Loan Management System — Technical Overview

Bank-grade lending for Bangladesh scheduled banks · Staff web · Borrower app · Field app · Fineract core

16 modules · 162 REST endpoints · 18 migrations

193 Java tests · 44 e2e journeys · 12 runbooks

What this deck covers

01

System at a glance

KPIs, scope, architecture

02

Technology stack

Binding stack v3 — every layer

03

Modules & functions

All 16 backend modules

04

Regulatory engine

BRPD 15/2024 · ECL · Basel · AML

05

Quality gates

311 tests across five suites

06

The three apps

Staff web · borrower · field

07

Deploy · enhance · secure

Environments, extension recipe, controls

08

Boundaries & roadmap

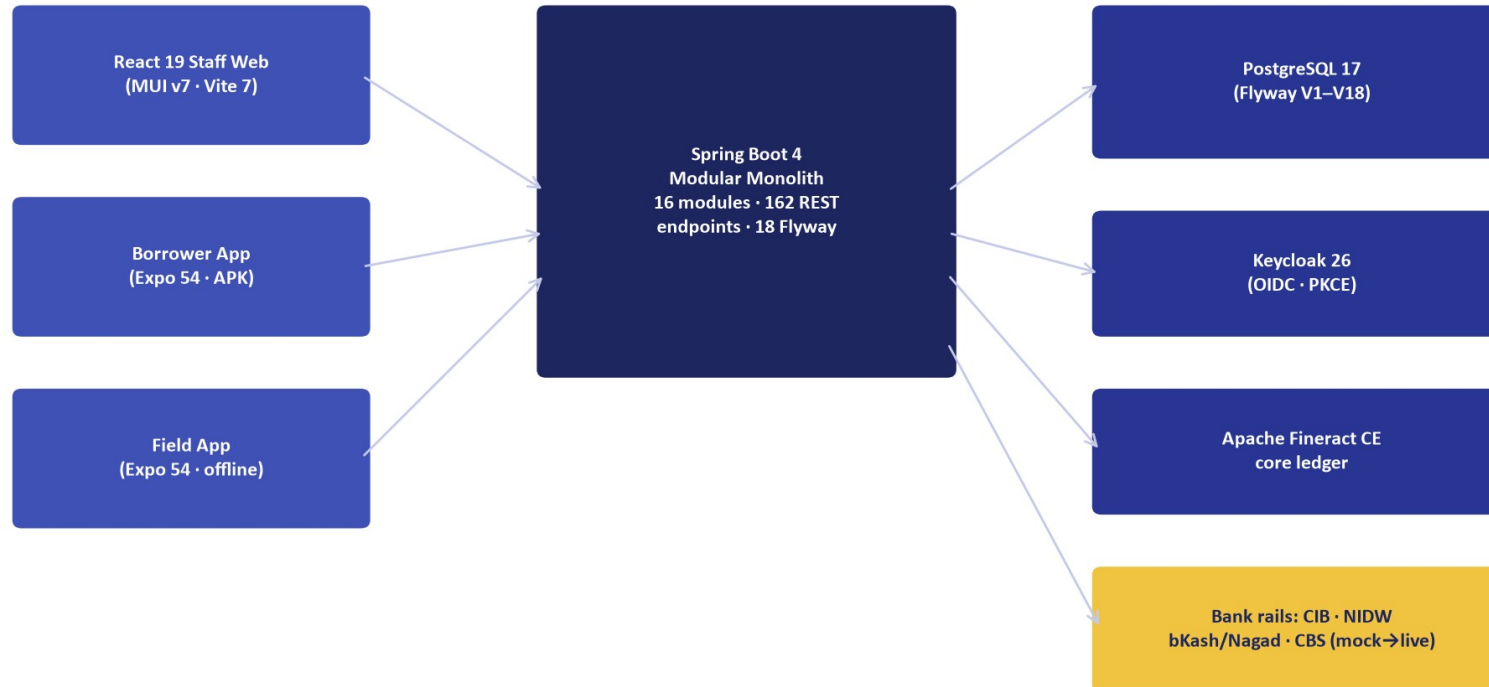
What is externally bound, what is next

The program in seven numbers



Modular monolith by binding decision — one deployable unit, module boundaries enforced by Spring Modulith. Apache Fineract CE is the immutable loan subledger; ULMS mirrors state for BRPD boards and projections. Build stage complete; bank cutover items follow the RB-11A flip runbook.

Three clients · one modular monolith · bank cores



Key decisions

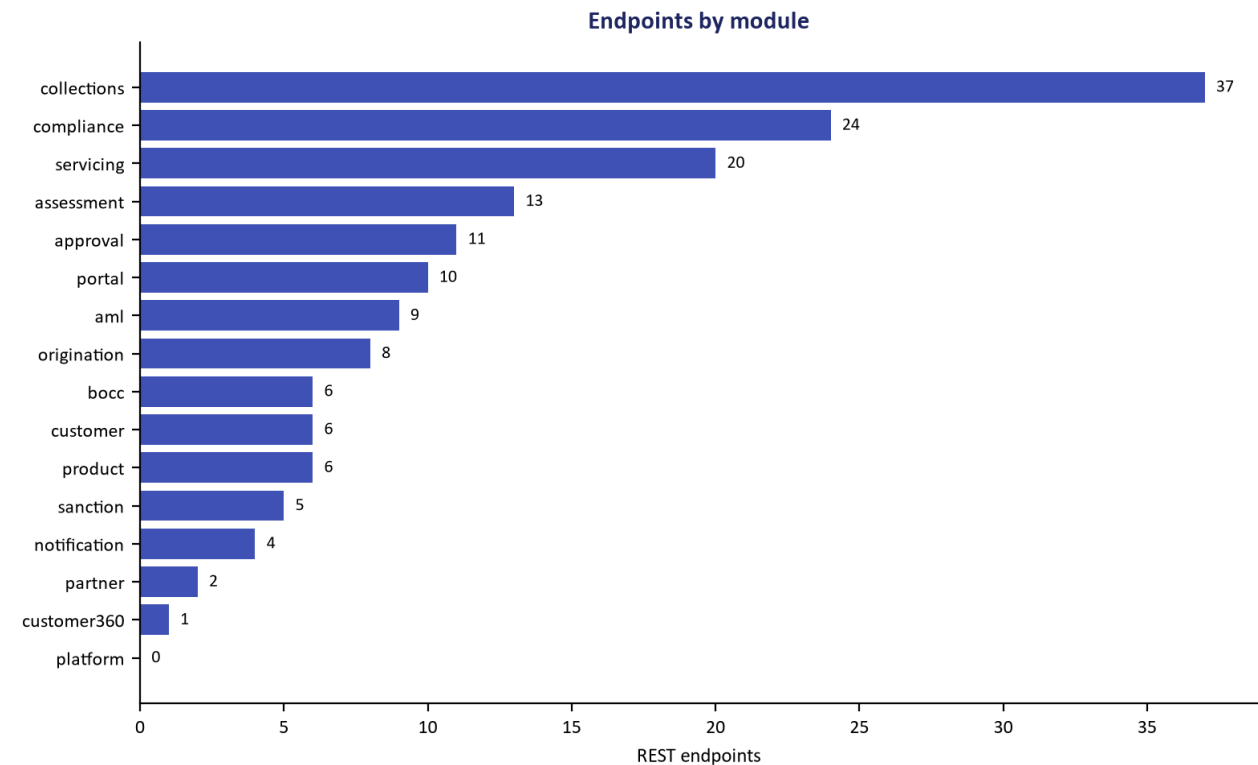
- ▶ Modular monolith (stack v3) — cycles broken with owned ports
- ▶ Contract-first mock drives 44 e2e tests without Docker
- ▶ Live adapters profile-gated (cib-live, rails-live ...)
- ▶ nginx same-origin: /api, /realms (Keycloak), /hooks

Source: component diagram generated from repo structure (SOURCES.md, Marketing/exhibits/E1).

Binding Technology Stack v3 — every layer

Layer	Technology	Notes
Backend	Java 21 · Spring Boot 4 · Gradle 8	Modular monolith · Spring Modulith
Database	PostgreSQL 17 · Flyway	18 migrations V1–V18 · JPA validates only
Identity	Keycloak 26 · OIDC + PKCE	Realm ulms · direct-grant MFA login
Staff web	React 19 · MUI v7 · Vite 7 · TS 5.8	Dynamics-365 shell · bilingual EN/BN
Mobile	Expo 54 · React Native 0.81	Borrower (signed APK) · field (offline-first)
Core ledger	Apache Fineract CE (pinned SHA)	Loan lifecycle subledger
Ops	Compose → k3s · Helm · Prometheus/Grafana	6 CI stages · 14 jobs (GitLab)
Contract	OpenAPI 3.0 (redocly-valid)	140 documented paths

16 modules — where the endpoints live



37 Collections **24** Compliance

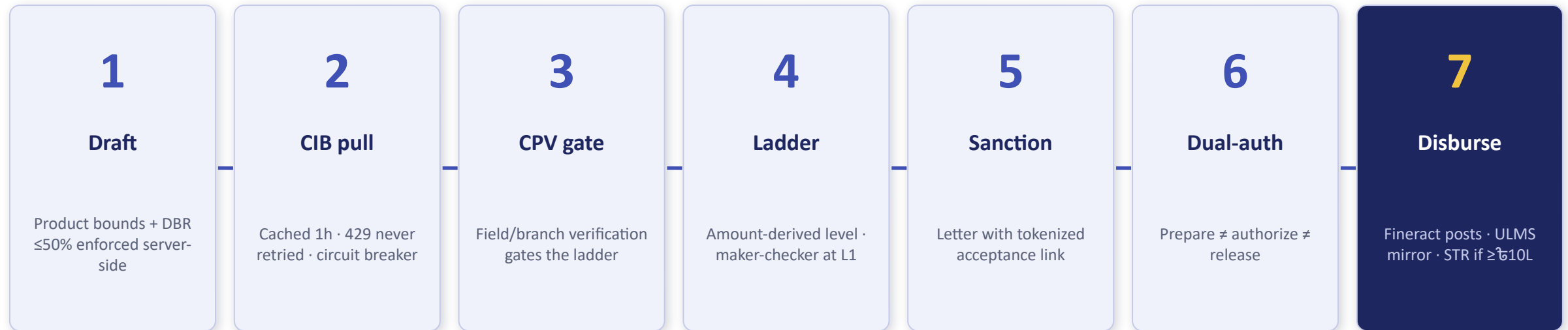
20 Servicing **13** Assessment

11 Approval **10** Portal

9 AML **8** Origination

Counts from the structured repo sweep (SOURCES.md). Collections & compliance dominate by design — delinquency and regulatory reporting are the deepest domains.

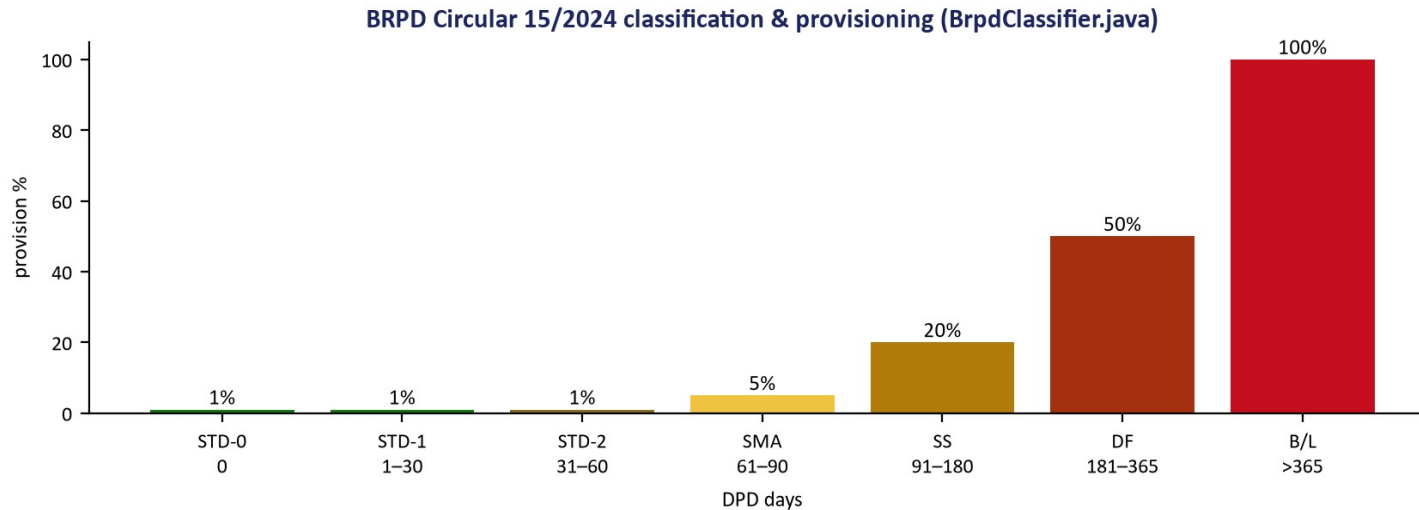
Credit-to-cash: application to money in seven steps



Nightly EOD (23:30 Dhaka) then reclassifies the book: DPD recompute → BRPD stage → provision JV → interest suspense (SS+) → board & IFRS-9 refresh.

Source: workflow trace in code — OriginationService.java, ApprovalService.java, DisbursementService.java, EodBatchService.java.

BRPD Circular 15/2024 — classification & provisioning



Also implemented and tested

- ▶ IFRS-9 ECL staging (EclModel.java)
- ▶ Basel III CAR + RWA by segment + parallel-run harness
- ▶ BFIU AML: STR ≥ 20L cash, CTR, goAML XML export
- ▶ 12-return Regcon catalog · WORM packs · maker-checker sign-off

Interest suspense applies from SS onward; ≥EMI payments regularize DPD (demoed live in the borrower app demo mode). Boundaries locked by BrpdBoundaryLockTest — DF ≤365 / B-L >365 move only together.

Source: BrpdClassifier.java, EclModel.java, BaselParallelRunService.java, AmlController.java.

Seven levels, configured — not coded

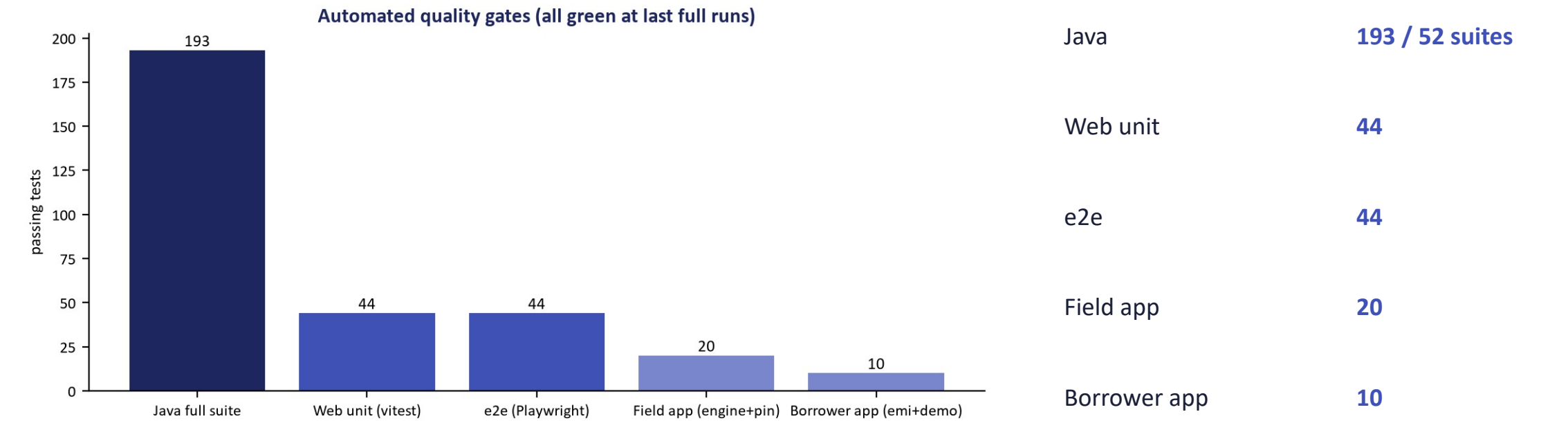
7-level maker-checker approval ladder (V2 approval_band seed)



L1 nodes are maker-checker dual; SLA 48h/level with 02:15 nightly auto-escalation. DELEGATE moves a task to a peer (assignee-only gate after); APPROVE_WITH_CONDITIONS records conditions precedent that block disbursement until SATISFIED or WAIVED. Bands live in approval_band (V2 seed) — the bank retunes limits with a data change.

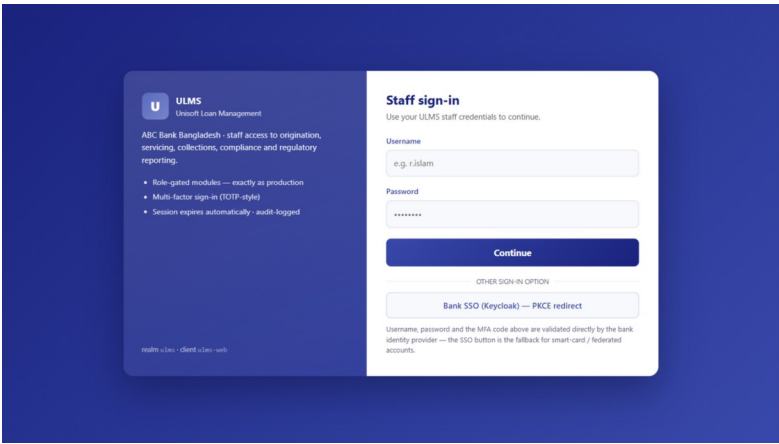
Source: V2__origination_workflow.sql:98-105, SlaService.java, Q1DelegateConditionsTest.java.

311 passing tests across five suites

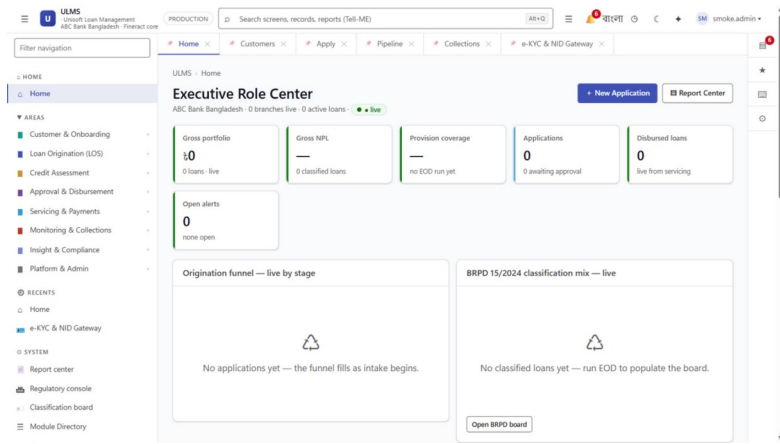


Java suites run in Docker (gradle:8-jdk21 + Testcontainers PG17 — CI parity). The e2e suite includes a WCAG 2.1 AA axe gate over 7 surfaces; the build reproduced the release APK byte-identically (deterministic pipeline).

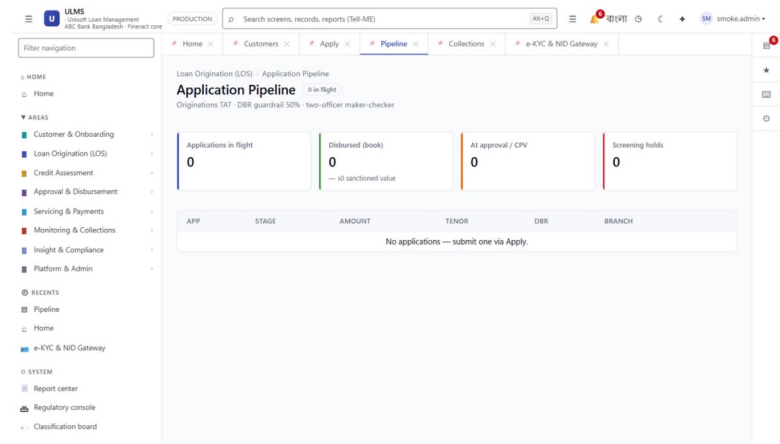
Production staff cockpit — real Keycloak session



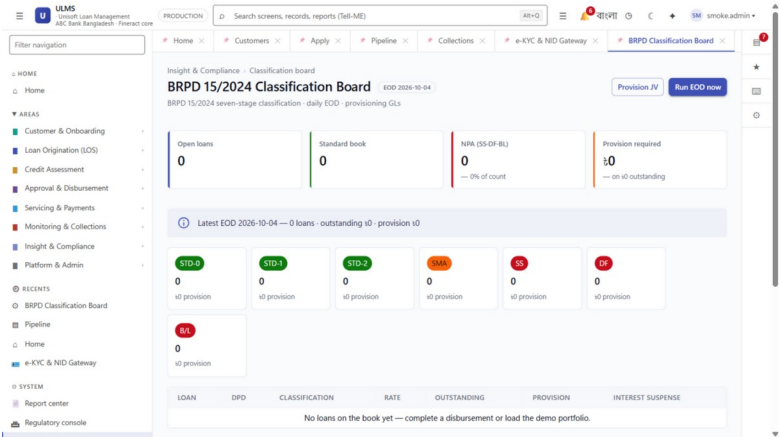
Sign-in: creds → MFA (Keycloak direct grant)



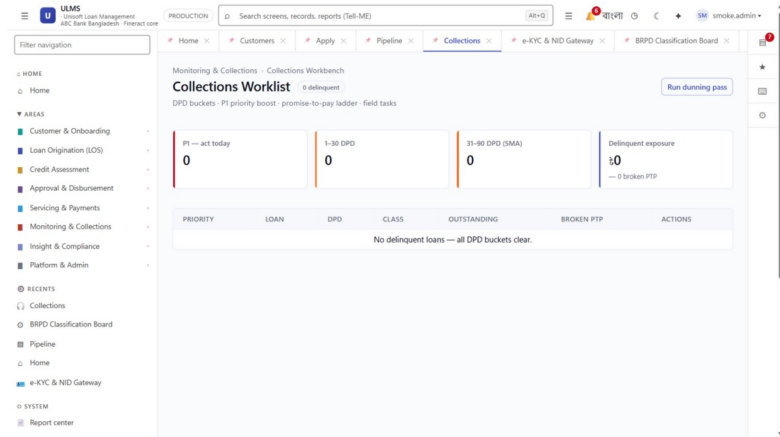
Home — environment chip reads PRODUCTION



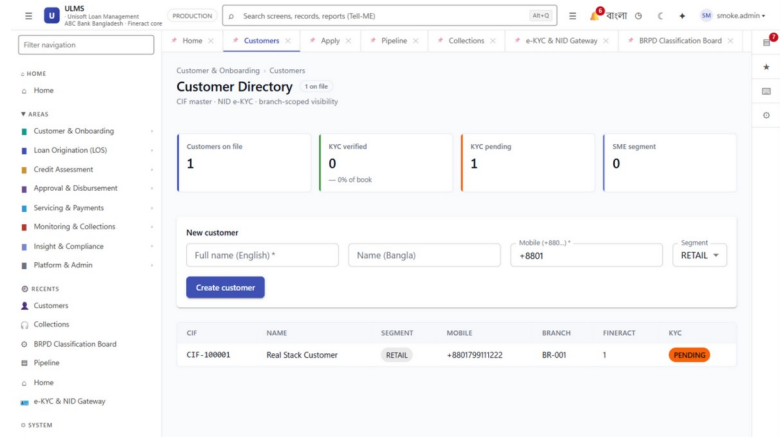
Pipeline — stage chips, DBR, drawer



BRPD classification board

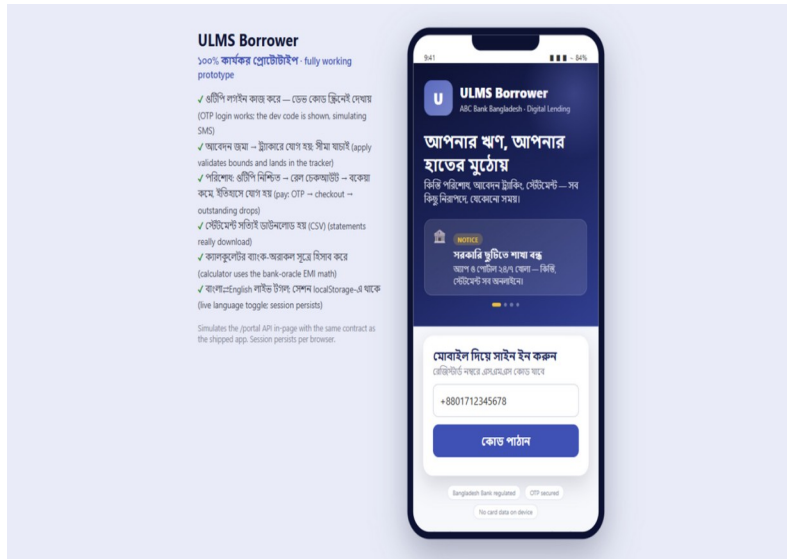


Collections workbench — PTP, dunning

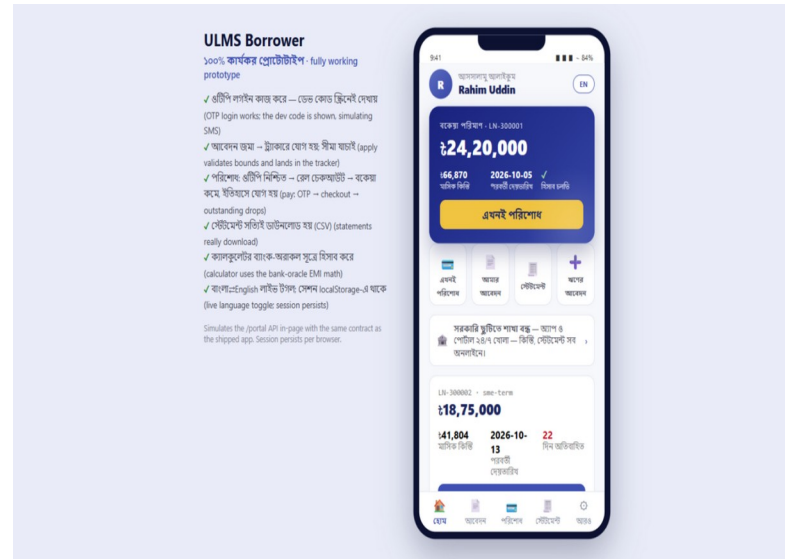


Customer master — branch-scoped

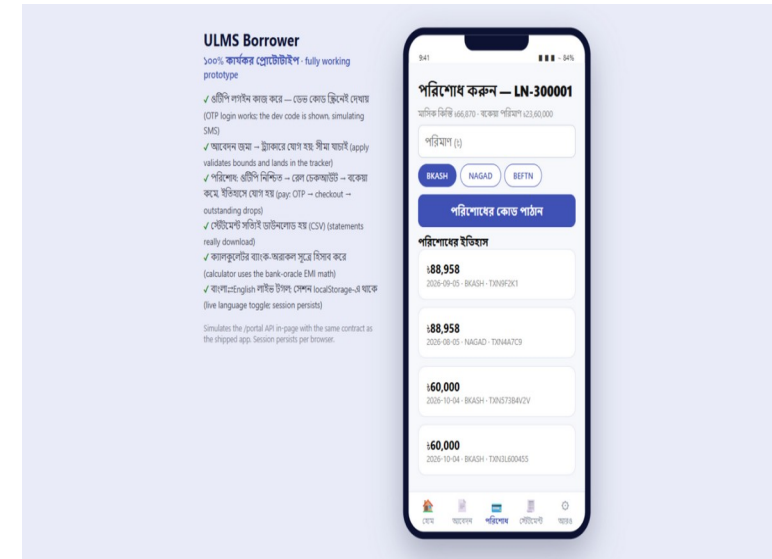
Signed Android APK — runs standalone in demo mode



OTP sign-in — code auto-fills (demo simulates SMS)



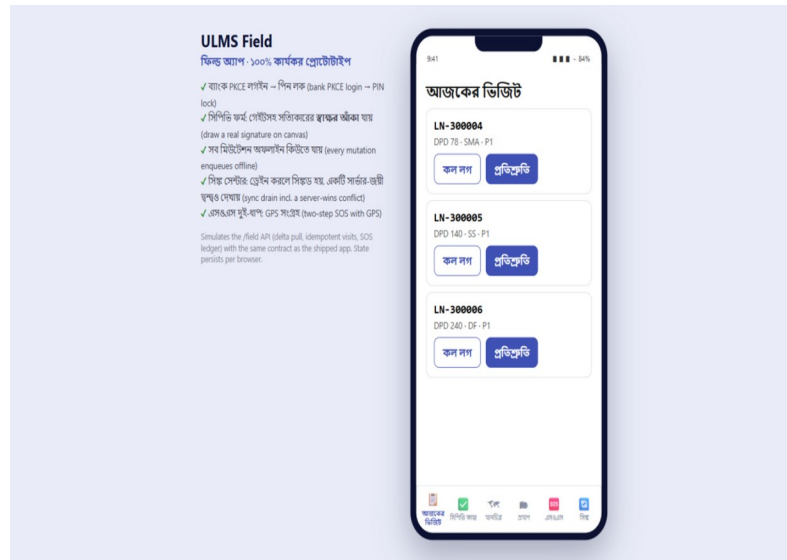
Home — hero balance card, EMI, DPD, quick actions



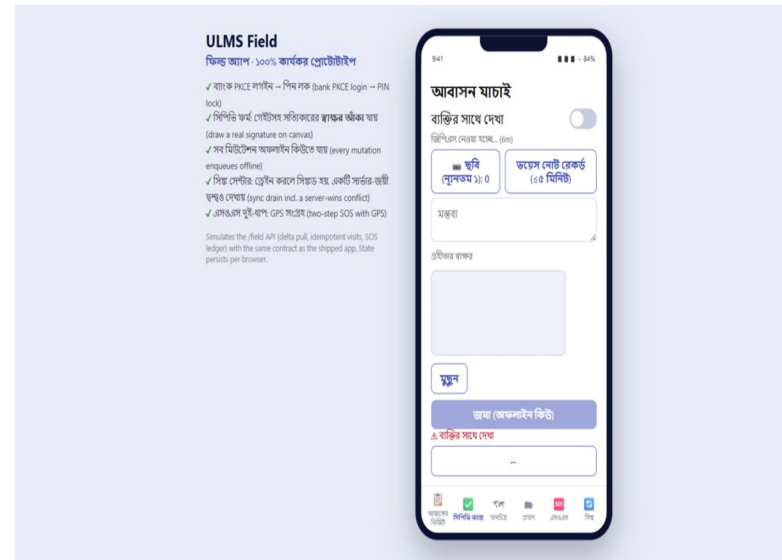
Pay — rail chips + OTP confirmation

- ▶ com.uslbd.ulmsborrower v0.1.0 · minSdk 24 · apksigner v2/v3 verified · arm64 Hermes
- ▶ No backend set → embedded prototype simulation; set EXPO_PUBLIC_API_BASE → real /portal
- ▶ iOS: verified Xcode project shipped — IPA needs Mac/EAS per Apple licensing

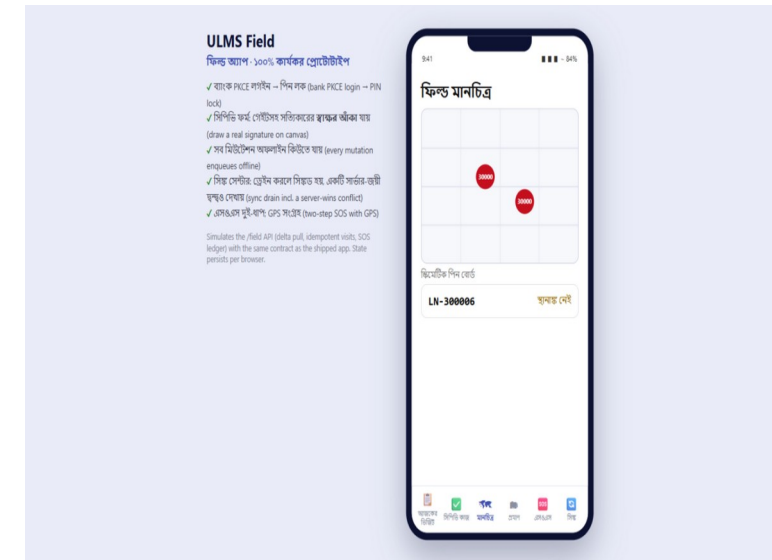
Offline-first verification — evidence never lost



Bank PKCE sign-in + security notices



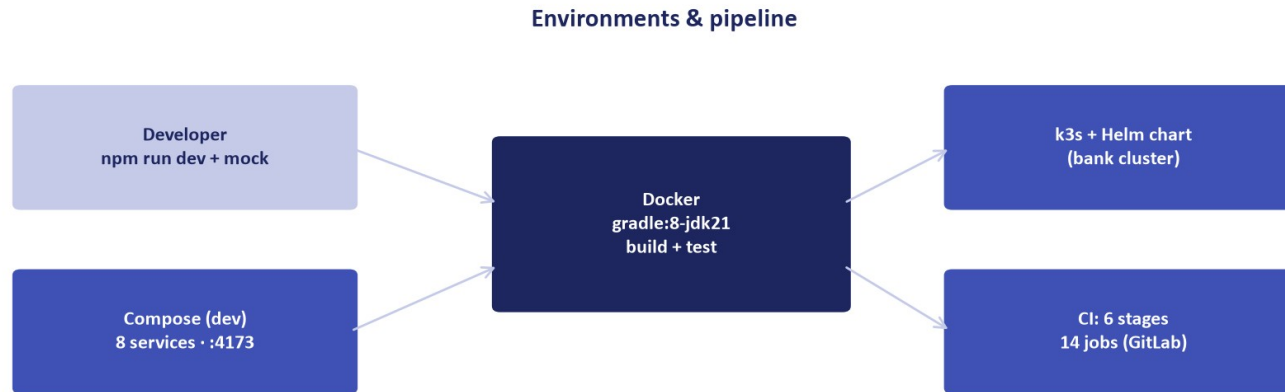
CPV form — signature canvas, GPS gate, voice ≤5 min



Field map — borrower task pins

- ▶ Visits idempotent by clientId · task state is server-wins · evidence append-only
- ▶ App-PIN lock (salted hash) + two-step SOS that never silently drops
- ▶ Dedicated /field gateway: delta pull with bundleVersion, visits, PTP, SOS + ack desk

Three environments, one pipeline



Get running

```
npm run mock:api && npm run dev    # zero-Docker dev
docker compose up -d                # 8-service
verification
helm upgrade --install ulms deploy/chart/ulms --atomic
```

- ▶ 10 scheduled jobs (SLA */15 · EOD 23:30 · WORM 02:00 · dunning 06:00 ...)
- ▶ 12 runbooks incl. RB-11A flip matrix — mock→bank is values-only
- ▶ Mobile: build-apk.sh one command → signed APK (byte-identical rebuild proven)

Controls a regulator can trace

Identity

Keycloak OIDC · PKCE · MFA on staff login · SSO secondary

Authorization

Role gates per endpoint · ladder-level gates · own-CIF portal rows

Dual control

Maker-checker ladder · triple-gated disbursement · SOS ledger

Audit

WORM audit_entry on every mutation · signatures on approvals

Secrets

Env/Vault only — zero usable literals in the repo (scanned)

Regulatory

BRPD · IFRS-9 · Basel III · BFIU AML (STR/CTR/goAML) · 12 returns

Plus: CSP + nosniff + XFO on the web tier · WCAG axe gate in CI · bilingual incl. Bangla numerals.

The six-step vertical-slice recipe

1

Entity + Flyway V19+

JPA validates; never generates

2

Repository + service

Rules + @PreAuthorize + Testcontainers test

3

Controller + OpenAPI

redocly must stay valid

4

Mock route

Contract parity for Docker-free e2e

5

Clients + screens

Web API client · mobile where needed

6

Suites green

Touched suites rerun in Docker

Reference slices in repo: collections watchlist + auctions (V17) and the field gateway (V18) each added table → service → controller → spec → mock → screen in one commit series.

Stated plainly — nothing hidden

Externally bound

CIB/NIDW/rail/CBS credentials · clustered Keycloak · bank CI runner · EAS/MDM signing. Mechanism: values-only RB-11A flip.

Deferred by decision

Multi-tenancy (D3) · Islamic Murabaha (phase 4) · outbox→Kafka · websockets. Revisit triggers documented.

UX-contract scope

~120 prototype archetype screens remain badged reference layouts; conversion = the analytics workstream.

Environment

Local Docker Desktop is crash-prone under build load — always-on demo should move to the bank k3s cluster.

Build stage: complete.

Deployment execution and bank-bound cutover follow the go-live checklist and the RB-11A flip runbook.

- ▶ Full technical documentation: `ULMS_v2_Technical_Documentation.docx` (34 tables, 18 exhibits, provenance map)
- ▶ Every claim traceable via `SOURCES.md` — re-verify in minutes
- ▶ Working prototypes: `localhost:8791/prototype` (borrower) · `/prototype/field` (field ops)